

Repair, Don’t Replan: A Controlled Study of Observation-Conditioned Remasking in Masked-Diffusion Planners

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Abstract

Masked-diffusion language models (dLLMs) can in principle edit a plan *in place*: when a tool observation invalidates part of a multi-step plan, the denoiser can remask exactly the affected tokens and re-denoise them jointly with the surviving plan. Existing remasking methods are *introspective* — they revise tokens the model itself judges low-quality — whereas an agent must revise tokens the *environment* has invalidated. We isolate this setting in a controlled testbed — 8-step tool plans, dataflow arguments, state-diff success, mid-episode perturbations, small masked-diffusion and autoregressive (AR) planners trained from scratch — in which the ground-truth invalidation set is computable. Three findings. (i) At matched success on in-distribution repairs, selective remasking regenerates 1.9 tokens per episode versus 83–84 for full or AR-suffix replanning, using 27 versus 108 network calls; a posterior-rescoring masker matches the oracle repair almost exactly (precision 0.92, recall 0.90). (ii) On compositionally novel perturbation pairs held out from training, every learned agent degrades (0.75–0.88) while the *same* denoisers with oracle masking retain 0.996: localization, not repair synthesis, is the bottleneck, and the failure concentrates entirely in repairs governed by an undemonstrated priority rule — and a controlled intervention closes it: adding just *eight* demonstrations of that rule to training repairs the composed conditional on every seed (0.41→0.997), where ten-fold more single-perturbation repair supervision does not. (iii) Outcome-reward RL restores repair from supervision-scarce SFT on every seed (→ 1.000) yet composes worse than demonstrations; the naive one-step policy gradient collapses, for a reason we identify. These are mechanism-study results on a synthetic environment with ~250k-parameter models, not a claim about 7–8B dLLMs.

1 Introduction

Diffusion language models are marketed on throughput, but a recent reality check finds they fail as agentic backbones: on AgentBoard and BFCL, LLaDA and Dream repeatedly retry actions instead of branching under temporal feedback and lose symbolic precision (e.g. strict JSON schemas) under diffusion noise, with embodied success rates below 2% [Lu et al., 2026, Nie et al., 2025, Ye et al., 2025]. That work names *diffusion-native agents* as the open direction. The strongest argument for such agents is not speed but **editability**: an AR model that discovers at step 2 that its step 6 is invalid can only append or regenerate a suffix, whereas a masked denoiser can return step 6 to [MASK] and re-denoise it jointly with the surviving plan, preserving valid *later* steps and their dataflow bindings.

Whether that promise is real is not obvious. “AR must replan” is a strawman: an AR model can be prompted to patch its plan, and prefix caching makes suffix regeneration cheap. Meanwhile most remasking work is introspective — RemeDi learns per-token confidence [Huang et al., 2025], ReMDM remasks on an inference-time schedule [Wang et al., 2025], CoRe flags context-brittle tokens [Zhai et al., 2026], with a rapid stream of follow-ups refining the same signal [Yao, 2026, Yu et al., 2026, Schiff et al., 2026, Hong et al., 2025], and Zhang et al. [2025] show that standard MDLM training does not even induce the ability to *recognize* incorrect visible tokens. Externally-triggered revision has begun to appear: Zou et al. [2026] repair outdated spans of a summary when the underlying document stream evolves. What remains unexamined — and what we are not aware

Workshop submission. All results are from the controlled synthetic testbed described in Section 2; see Section 1 for scope and Section 6 for limitations.



Figure 1: **Observation-conditioned selective repair.** A plan of 8 steps (3 tokens each) is drafted at turn 0; the refund code is left as a *typed hole* (dashed) because the tier is only revealed at step 3. Executing step 2 returns `SUSPENDED`, which invalidates step 6 only. The masker returns that step to `[MASK]` and the denoiser re-denoises it jointly with the intact plan. Full replanning would remask all 18 downstream tokens; an AR agent would regenerate the suffix. The showcase in Family B (Section 2) is harder still: steps 3 and 7 change while step 4–6 stay valid.

of prior work evaluating — is observation-conditioned remasking for *temporally unfolding tool-use plans*, where each executed step returns feedback, repairs must respect dataflow across turns, and a ground-truth invalidation set exists to score localization itself. And at 7–8B scale the confounds (weak base policy, JSON validity, RL instability) are entangled enough that a negative end-to-end result would say little about the mechanism.

We therefore study the mechanism where it can be measured. We build **ToolFlow**, a synthetic tool-use environment with 8-step plans, dataflow variables, state-diff success, and perturbations revealed by observations mid-episode — and, critically, a computable *oracle invalidation set*, which turns “how good is the masker?” into a precision/recall question and lets us separate repair *localization* from repair *synthesis*. We train small masked-diffusion and AR planners from scratch (identical architecture, bidirectional vs. causal), and compare five repair strategies plus an AR-patch baseline across 5 seeds.

Contributions. (1) A controlled testbed for observation-conditioned plan repair with an oracle invalidation set and a *compositional* generalization split: training episodes contain at most one perturbation, evaluation includes never-demonstrated perturbation *pairs*. (2) A head-to-head of repair strategies under matched success, with masker precision/recall, showing selective remasking obtains oracle-sized repairs (1.9 tokens) at $\sim 44\times$ less regeneration than replanning and $4\times$ fewer network calls than an AR agent. (3) The central diagnostic: with oracle masking the same denoisers execute novel composed repairs at 0.996, so the compositional gap is localization; per-family analysis pins the entire failure on repairs governed by a priority rule absent from training; a supervision-dose intervention (Appendix D) shows eight demonstrations of that rule suffice to fix it while ten-fold more single-perturbation supervision does not. (4) A post-training study on a supervision-scarce start with the controls the claim requires: a continued-SFT arm shows most of the gain is supervised, an isolated negative-advantage arm reproduces the collapse of the naive one-step estimator, and we correct our own earlier mechanistic account (Appendix C).

Scope. Every number here comes from a synthetic environment and $\sim 250\text{k}$ -parameter models trained from scratch on one CPU core. We make no claim of quantitative transfer to 7–8B dLLMs on Agent-Diff [Pysklo et al., 2026] or BFCL [Patil et al., 2025]. We claim the mechanism is implementable, its efficiency advantage is real under matched success here, and its failure mode is structured and diagnosable. Section 6 states what does not follow.

2 ToolFlow: a controlled testbed for plan repair

Plans and execution. A plan is $8 \text{ steps} \times 3 \text{ tokens}$ (`TOOL`, `ARG1`, `ARG2`) over a 22-tool library, written into a fixed 38-token sequence together with the task context and one observation slot per step. Arguments are either literals or *dataflow variables* `$k` referring to the binding produced by step `k`; a call whose argument resolves to the wrong binding type is rejected by the environment, increments an invalid counter, and returns `ERR`. Success is *state-diff*: the goal effect set must be a subset of

the effects achieved, as in Agent-Diff [Pysklo et al., 2026], so a plan is judged by what it changed in the world, not by matching a reference trace.

Families and perturbations. Three families (refund, support, RMA) each admit perturbations revealed only by an observation: e.g. a suspended account (revealed at step 2) forces step 6 from refund to escalate; a ticket-system outage forces steps 3 and 7 to change while step 4–6 remain valid (a non-contiguous, long-range repair that suffix regeneration cannot preserve); an out-of-warranty status and a premium tier independently force steps 3 and 5. Family A additionally carries a *typed hole*: the refund code depends on an order tier revealed only at step 3, so the token is unresolvable at draft time. Family A’s two perturbations *conflict* and are resolved by a dominance rule (suspension overrides open-invoice); Family C’s two *compose independently*.

Compositional split. Training instances carry **at most one** perturbation. Episodes with two ($n=2$) appear only at evaluation, so every $n=2$ number is compositional generalization by construction: the repair *primitives* are demonstrated, their *combination* never is.

Oracle invalidation. Because the environment knows the correct plan under the revealed information, we can compute, at each turn, exactly which committed future tokens contradict what is now known. This gives (a) an upper-bound agent (*oracle masker*) and (b) precision/recall for any learned masker — the measurement that makes the localization-vs-synthesis question answerable.

Models. A 3-layer, $d=96$, 4-head Transformer ($\sim 250k$ parameters) in JAX trained from scratch on expert snapshots: a masked-diffusion loss (mask rate $u \sim \mathcal{U}(0.15, 1)$ over future-step positions) for the denoiser, and the identical architecture with a causal mask and next-token loss for the AR baseline (2,000 steps, batch 64, AdamW, 5 seeds; Appendix E).

3 Agents

All diffusion agents share one denoiser and differ only in the **invalidation masker** applied after each observation. Let x be the sequence, C the set of committed future plan tokens after turn k . A masker M selects $S \subseteq C$; we set $x_S \leftarrow [\text{MASK}]$ and re-denoise S jointly (4 rounds, confidence-committing), leaving $x_{C \setminus S}$ untouched.

- **Fixed:** $S = \emptyset$ (no repair). **Full replan:** $S = C$ (remask all downstream).
- **Selective (oracle):** S = ground-truth invalidated tokens (upper bound).
- **Selective (conf.):** *posterior rescoring*. For each future step j , mask its 3-token slot and read the denoiser’s posterior of the *currently committed* token under the updated context; remask tokens with $p < \tau_{\text{inv}}=0.35$. One batched forward pass over the (episode $\times$ slot) grid per turn.
- **AR replan:** regenerate the remaining suffix after each observation.
- **AR patch:** the AR analogue of selective repair — score each committed future token under the updated prefix, flag $p < \tau_{\text{inv}}$, and locally resample flagged positions left to right. Its NFE is counted optimistically (idealized prefix-cache reuse).

Typed holes. On the final denoising round, positions whose max probability falls below $\tau_{\text{hole}}=0.62$ stay masked rather than being guessed, and are filled just-in-time before their step executes. This implements *defer-until-observed* for tokens that are genuinely unresolvable at draft time (the refund code).

Post-training. We post-train the denoiser from rollouts: 8 tasks $\times$ 6 rollouts, reward $2 \cdot \text{success} - 0.2 \cdot \text{invalid} - 0.3 \cdot (\text{regen}/24)$, group-relative advantages [Shao et al., 2024], and the one-step masked-likelihood estimator used by dLLM RL methods [Zhao et al., 2025, Zhu et al., 2025] — which, in the decomposition of Raajesh et al. [2026], optimizes only the *token* term of the exact MDLM policy gradient and omits the masking term. The stable recipe clips advantages to $[0, 1.5]$, i.e. imitates above-mean rollouts; we therefore call it **reward-filtered imitation** (RAFT-style, Dong et al., 2023) rather than a policy gradient. Headroom comes from a deliberately compute-limited SFT start (425 steps, $\sim 21\%$ budget, repair supervision at 10% coverage). Appendix C gives losses, controls, and diagnostics.

Table 1: **Main results** (5 training seeds, 400 episodes/cell, mean $\pm$ std over seeds). n = perturbations per episode; $n=2$ combinations are held out from training. Regen./NFE/Invalid are per episode at $n=1$. Masker P/R is against the oracle invalidation set. All repair-capable agents match on success at $n=1$; they differ by $\sim 44\times$ in regeneration.

Agent	Task success $\uparrow$			Cost / errors at $n=1$ $\downarrow$			Masker P / R ($n=2$)
	$n=0$	$n=1$	$n=2$ (novel)	Regen.	NFE	Invalid	
Fixed plan (no repair)	0.922 $\pm$ 0.044	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.0 $\pm$ 0.0	7 $\pm$ 0	2.02 $\pm$ 0.05	–
Full replan	0.963 $\pm$ 0.057	0.966 $\pm$ 0.067	0.764 $\pm$ 0.210	83.3 $\pm$ 0.3	33 $\pm$ 0	0.01 $\pm$ 0.02	–
AR replan (suffix)	0.989 $\pm$ 0.021	0.998 $\pm$ 0.004	0.754 $\pm$ 0.155	84.0 $\pm$ 0.0	108 $\pm$ 0	0.00 $\pm$ 0.01	–
AR patch	0.993 $\pm$ 0.015	1.000 $\pm$ 0.001	0.884 $\pm$ 0.211	2.0 $\pm$ 0.1	33 $\pm$ 0	0.00 $\pm$ 0.00	0.96 / 0.64
Selective (conf.)	0.977 $\pm$ 0.045	0.974 $\pm$ 0.052	0.785 $\pm$ 0.239	1.9 $\pm$ 0.1	27 $\pm$ 4	0.00 $\pm$ 0.01	0.92 / 0.90
Selective (oracle)	0.982 $\pm$ 0.036	0.977 $\pm$ 0.046	0.996 $\pm$ 0.006	1.9 $\pm$ 0.1	20 $\pm$ 4	0.00 $\pm$ 0.00	–
<i>Supervision-scarce condition (5 seeds): selective (conf.) agent, 425-step SFT</i>							
lite-SFT	0.898 $\pm$ 0.056	0.838 $\pm$ 0.062	0.365 $\pm$ 0.230	1.8 $\pm$ 0.1	31 $\pm$ 3	0.20 $\pm$ 0.08	–
lite-SFT + RL	0.932 $\pm$ 0.067	0.996 $\pm$ 0.003	0.401 $\pm$ 0.361	1.8 $\pm$ 0.1	27 $\pm$ 1	0.01 $\pm$ 0.01	–

Metrics. Task success (state-diff); **regenerated tokens** (committed tokens written again — our efficiency headline, since wall-clock against optimized AR serving measures the serving stack, not the mechanism); invalid calls; and **NFE** (network forward passes). 400 episodes per cell, decode temperature 0.7.

4 Results

Repair is necessary and cheap. Without repair, success at $n \geq 1$ is exactly zero and the agent issues 2.02 invalid calls per episode (Table 1) — the perturbations bite. Every repair-capable agent recovers to 0.97–1.00 at $n=1$, so *success does not separate them*; cost does. Selective remasking regenerates 1.9 tokens per episode against 83.3 (full replan) and 84.0 (AR suffix), a $\sim 44\times$ reduction, and needs 27 network calls against 108 for the AR agent (Figure 2). The learned masker’s repairs are oracle-sized (1.9 vs. 1.9), and on unperturbed episodes it is nearly silent (0.1 regenerated tokens), so selectivity costs nothing when nothing is wrong.

The learned masker is oracle-grade in-distribution. Posterior rescoring reaches precision 0.92 / recall 0.90 against the ground-truth invalidation set, versus 0.96/0.64 for AR prefix-scored patching (Table 1). The recall gap is the expected structural difference: a bidirectional denoiser scores a committed token against the *whole* sequence including the observation and the downstream plan, while an AR prefix score cannot see what comes after. The masker is also insensitive to its one hyperparameter: sweeping τ_{inv} over $[0.10, 0.65]$ moves $n=1$ success by less than 0.01 and trades precision for recall only mildly, and *no* threshold rescues composition — $n=2$ success is flat across the whole range (Appendix B, Figure 6), confirming the failure lives in the conditional, not the operating point.

AR patch is a strong baseline, and we report it as such. With local editing rather than suffix regeneration, the AR agent matches selective repair on success (1.000 vs. 0.974 at $n=1$) and on regenerated tokens (2.0 vs. 1.9). The honest diffusion-side advantages that survive this comparison are narrower than the usual pitch: fewer network calls (27 vs. 33, and the AR count is idealized), higher masker recall, and joint multi-slot re-denoising. Any claim that in-place editing is diffusion’s exclusive agentic advantage does not survive contact with this baseline.

Localization, not synthesis, is the compositional bottleneck. On never-demonstrated perturbation pairs, every learned agent drops (full 0.764, AR 0.754, selective 0.785, AR patch 0.884) while the oracle-masked agent — *the same denoisers, only the masker replaced* — holds 0.996 ± 0.006 (Table 1). The denoisers can therefore execute composed repairs they never saw; they cannot reliably *detect* them. Figure 3 localizes the failure completely: on independently-composing doubles (Family C) every agent succeeds (≥ 0.988), whereas on conflicting doubles (Family A), resolved by a dominance rule absent from single-perturbation training, per-seed success is bimodal — each model essentially commits to one resolution and is right or wrong for all episodes (selective 0.585,

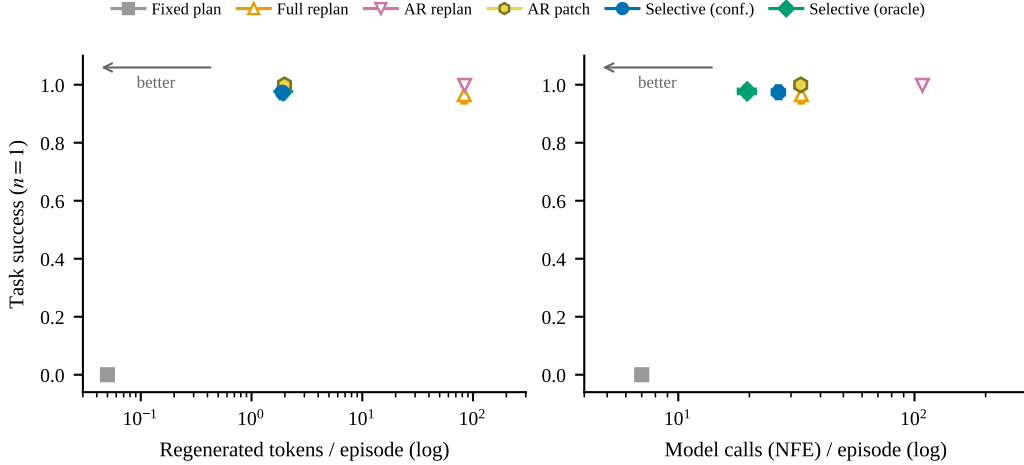


Figure 2: **Cost at matched success** ($n=1$, 5 seeds, log x). Repair-capable agents are indistinguishable on success and separated by 1.5 orders of magnitude in regenerated tokens (left) and $4\times$ in network calls (right). Error bars are s.e.m. over seeds.

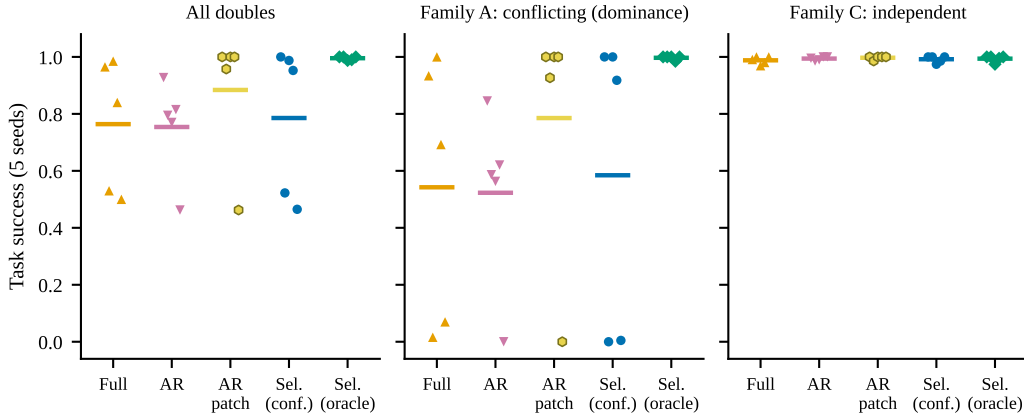


Figure 3: **Localization is the compositional bottleneck.** Success on held-out double-perturbation episodes; markers are the 5 seeds, bars are means. *Left*: all doubles. *Middle*: Family A, where the two perturbations conflict and are resolved by a dominance rule never demonstrated in training — every learned agent is bimodal across seeds (≈ 0 or ≈ 1). *Right*: Family C, where the two perturbations compose independently — every agent succeeds (≥ 0.988). The oracle-masked agent uses the *same* denoisers and is at 0.996 throughout, so the repairs themselves are synthesizable; what fails is deciding *what* to repair.

AR patch 0.785, full replan 0.542; per-seed values in Appendix A). Free regeneration does not help: full replanning, which rewrites everything, is the *worst* of the repair-capable agents here. A direct probe pins the mechanism: masking step 6 under the composed context (SUSPENDED + INV_OPEN) and reading the denoiser’s posterior gives $p(\text{escalate})$ of 0.85/0.05/0.96/0.62/0.16 across the five seeds — and whether this single conditional clears the invalidation threshold τ_{inv} predicts the end-to-end Family-A outcome on every seed. The oracle-masked agent partly wins by *never re-querying* this unreliable conditional: under dominance-aware localization, `escalate` is written once at the first reveal and never remasked. The implication — that repair *priority* rules must be represented in the supervision and do not emerge from demonstrations of the primitives — is testable, and we test it next.

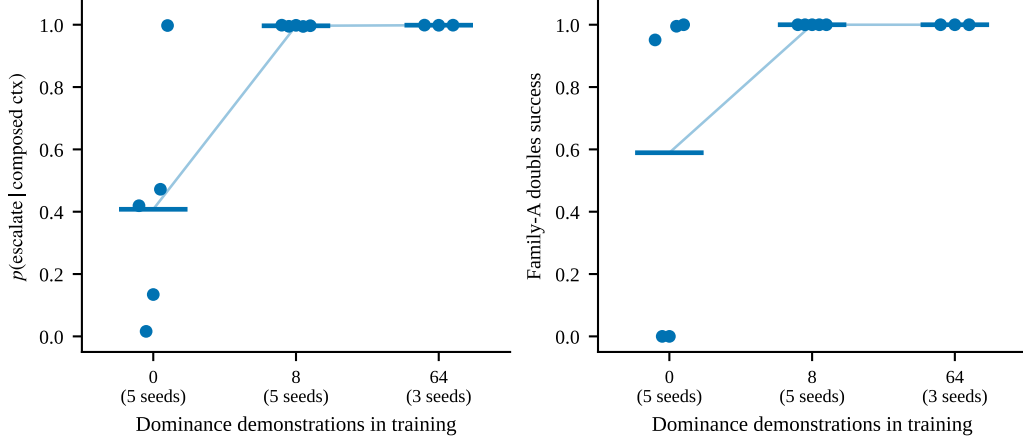


Figure 4: **Supervision-dose intervention** (self-contained cov = 1.0 grid; dots are seeds, bars are means). Left: the composed-context conditional $p(\text{escalate})$ that Figure 3 showed to be the failure point. Right: end-to-end Family-A doubles success. Eight demonstrations of the dominance rule move both from coin-flip to ceiling on every seed; 64 adds nothing further.

Eight demonstrations of the priority rule fix it. We add k training instances carrying both Family-A perturbations with the dominance-correct expert repair ($k \in \{0, 8, 64\}$; Family-C doubles remain fully held out), in a self-contained grid at full repair coverage (Appendix D). At $k=0$ — which already contains *ten times more* single-perturbation repair supervision than the base models — the composed conditional remains a coin flip: $p(\text{escalate} | \text{composed}) = 0.41 \pm 0.34$, with 1 of 5 seeds dominance-consistent. At $k=8$ it is 0.997 ± 0.002 with 5 of 5 seeds consistent (Fisher exact, one-sided $p = 0.024$), Family-A doubles success rises $0.59 \rightarrow 1.00$, and $k=64$ is saturated (Figure 4). A compute-only control makes the contrast sharp: continuing the two dominance-failing *base* models for a second full training budget leaves Family-A doubles at 0.000 and drives the composed conditional *further* from the rule ($0.05 \rightarrow 0.00$ and $0.16 \rightarrow 0.00$) while everything else stays perfect — extra compute entrenches whichever resolution the model drew. Neither volume of primitive supervision nor compute buys the priority rule; a handful of demonstrations of the rule itself does. Appendix D reports the grid’s side findings with the same candor, including a repair-corruption instability of the $k=0$ arm that the intervention also removes. The pattern is also *scale-stable*: across a $0.19 \times -2.1 \times$ parameter sweep and a $4 \times$ -data arm, dose-0 draws are dominance-consistent in 3 of 13 cases while every healthy dose-8 draw is (10 of 10; exploratory pooled Fisher $p = 2.5 \times 10^{-4}$; Appendix D, Figure 10).

Regeneration exposure grows with decode noise. Sweeping decode temperature (Appendix B, Figure 5) preserves the ordering and widens the gaps: at $T=1.1$ on in-distribution repairs, full replan falls to 0.929 while selective holds 0.958 and AR patch 0.982; on novel doubles the oracle-masked agent still reaches 0.936 where full replan reaches 0.716. Rewriting correct tokens is not free — each rewrite is another chance to sample a wrong one.

Typed holes. Deferring unresolvable tokens until the relevant observation raises unperturbed success from 0.823 ± 0.020 to 0.921 ± 0.057 and halves invalid calls ($0.177 \rightarrow 0.079$) for an agent with *no* revision mechanism at all (Figure 9). One seed of five reverses the effect; we plot all seeds rather than the mean alone.

Post-training restores repair; the reward deserves little of the credit, and composition none. From the supervision-scarce checkpoint, reward-filtered imitation lifts probe success from 0.71–0.91 to 0.99–1.00 on all five seeds in 200 iterations; at $n=1$ evaluation, $0.838 \pm 0.062 \rightarrow 0.996 \pm 0.003$ with invalid calls $0.20 \rightarrow 0.01$ and minimal regeneration (Table 1, Figure 7). Two caveats keep this honest. First, a continued-SFT control (the replay anchor alone, no reward signal) reaches 0.979 on the same probe — most of the gain is supervised learning, not reward (Appendix C). Second, the

recovered skill does not compose: on novel doubles the post-trained policy scores 0.401 ± 0.361 (per-seed 0.52/0.99/0.44/0.02/0.03) versus 0.785 for demonstration-trained models. We had initially attributed the collapse of the naive signed-advantage estimator to a “saturation asymmetry”; the derivation and controlled experiments in Appendix C show that account was wrong as stated, isolate the negative-advantage direction as the destructive component, and bound what we can and cannot conclude.

5 Related work

Remasking and correction in dLLMs. The introspective line detects the model’s own errors: RemeDi learns per-token confidence with SFT+RL [Huang et al., 2025]; ReMDM provides an inference-time remasking sampler [Wang et al., 2025]; CoRe flags context-brittle tokens [Zhai et al., 2026]; further work refines detection or revises without remasking [Yao, 2026, Yu et al., 2026, Schiff et al., 2026, Hong et al., 2025]. Two papers are closer to us. Zhang et al. [2025] build a controlled executable benchmark for error localization and in-place program correction, showing standard MDLM objectives give visible tokens no gradient signal and hence no error-awareness; our slot-masked rescorer sidesteps exactly that limitation by re-masking a slot before scoring it, and our dominance result echoes their conclusion that the needed discrimination must be put into training. Zou et al. [2026] repair outdated summary spans when an external document stream evolves, with a trained span detector; relative to both, our setting is a temporally unfolding tool-use loop with executable state, dataflow arguments, and a ground-truth invalidation set that lets us score localization directly. **RL for dLLMs** — d1/diffu-GRPO [Zhao et al., 2025], VRPO [Zhu et al., 2025], DiffuCoder [Gong et al., 2025], wd1 [Tang et al., 2025] — optimizes single-response reasoning with one-step likelihood surrogates; Raajesh et al. [2026] formalize the exact policy gradient as a two-stage action MDP whose token term these surrogates keep and whose masking term they drop. We reuse the token-term estimator in a multi-turn loop, report where it breaks, and decompose why (Appendix C). **Plan repair** is classical: our regeneration penalty is exactly *plan stability* [Fox et al., 2006], and AdaPlanner [Sun et al., 2023] repairs AR plans from feedback via prompting, which our AR-patch baseline instantiates; ReAct-style agents [Yao et al., 2023] avoid the problem by never committing to a plan. **Diffusion planners** in control replan by re-denoising trajectories [Janner et al., 2022, Chen et al., 2024], but with continuous states and no symbolic tool schema. **dLLM agents:** Lu et al. [2026] document the failure modes we take as motivation; Agent-Diff [Pysklo et al., 2026] supplies the state-diff evaluation contract we adopt in miniature.

6 Limitations and outlook

The environment is synthetic and the models are $\sim 250k$ parameters trained from scratch; nothing here establishes behavior of 7–8B dLLMs on free-form JSON plans and real APIs, where Lu et al. [2026] show symbolic precision itself is the binding constraint. Plans use fixed-length slots, so insertion/deletion repairs are out of scope (noop is the only deletion mechanism); block diffusion [Arriola et al., 2025] is the natural fix. Success is at ceiling at $n \leq 1$ for all repair-capable agents, so our claims there concern cost and stability, not success. Our AR-patch NFE is idealized in AR’s favor. One of five diffusion seeds converged to a visibly weaker model and is retained in every aggregate. The supervision-dose intervention deliberately breaks the strict held-out split for Family A (that is what an intervention is); Family C remains held out throughout, and the dose grid’s training configuration differs from the base models’ (Appendix D); its scale sweep spans 46k–526k parameters and is evidence about that range only, not about 7–8B models. The post-training study uses 5 seeds with the recipe tuned on one of them, optimizes only the token term of the exact MDLM policy gradient [Raajesh et al., 2026], and its ablation arms were run on a single seed.

The port is direct: swap the denoiser for LLaDA/Dream, keep `rescore_future` as one batched slot-masked forward per observation, replace ToolFlow with sandboxed enterprise APIs and a dataflow DSL under constrained decoding, and keep both the single-vs-composed split and the oracle-masker measurement — the dominance-conflict probe is directly replicable and is, on this evidence, where the interesting failure lives.

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A Per-family composition results

Table 2: Held-out double perturbations, split by family (5 seeds, 400 episodes/cell). Family A’s perturbations conflict (dominance rule, never demonstrated); Family C’s compose independently. Per-seed values expose the bimodality that the mean hides.

Agent	Family A: <i>conflicting</i> (dominance rule)		Family C: <i>independent</i>
	Success	per-seed	Success
Full replan	0.542 ± 0.421	[0.93, 0.02, 1.00, 0.69, 0.07]	0.988 ± 0.012
AR replan (suffix)	0.523 ± 0.280	[0.85, 0.59, 0.56, 0.62, 0.00]	0.994 ± 0.006
AR patch	0.785 ± 0.394	[1.00, 0.93, 1.00, 1.00, 0.00]	0.997 ± 0.006
Selective (conf.)	0.585 ± 0.476	[1.00, 0.00, 1.00, 0.92, 0.00]	0.992 ± 0.011
Selective (oracle)	0.997 ± 0.006	[1.00, 1.00, 1.00, 0.98, 1.00]	0.994 ± 0.010

B Additional figures

C What does the reward buy? Decomposing the post-training gains

Exact losses. Let a rollout produce a final sequence x with plan tokens y_i at positions i , and let S be a random mask over plan positions (rate 0.5), $\tilde{x}$ the sequence with S masked, and $\ell_i = \log p_\theta(y_i | \tilde{x})$ the one-step log-likelihood [Zhao et al., 2025]. With group-relative advantages $\hat{A}_b$ [Shao et al., 2024] over $G=6$ rollouts per task and an SFT replay anchor $\mathcal{L}_{\text{SFT}}$, the variants are

$$\mathcal{L}_{\text{naive}} = -\frac{1}{B} \sum_b \hat{A}_b \frac{1}{|S_b|} \sum_{i \in S_b} \ell_i + \beta \mathcal{L}_{\text{SFT}}, \quad (1)$$

$$\mathcal{L}_{\text{clip}} = -\frac{1}{\sum_b |S_b|} \sum_{b,i} \min(r_i A_b, \text{clip}(r_i, 1 \pm \epsilon) A_b) + \beta \mathcal{L}_{\text{SFT}}, \quad r_i = e^{\ell_i - \ell_i^{\text{old}}}, \quad (2)$$

with A_b a transform of $\hat{A}_b$. Our final recipe uses $A_b = \text{clip}(\hat{A}_b, 0, 1.5)$ — non-negative, hence *reward-filtered imitation* [Dong et al., 2023], not an unbiased policy gradient — with $\epsilon=0.2$, $\beta=1$, $\text{lr } 10^{-4}$. The historical v1 collapse used $\mathcal{L}_{\text{naive}}$ with $\text{lr } 5 \times 10^{-4}$, $\beta=0.3$, unclipped $\hat{A}_b$.

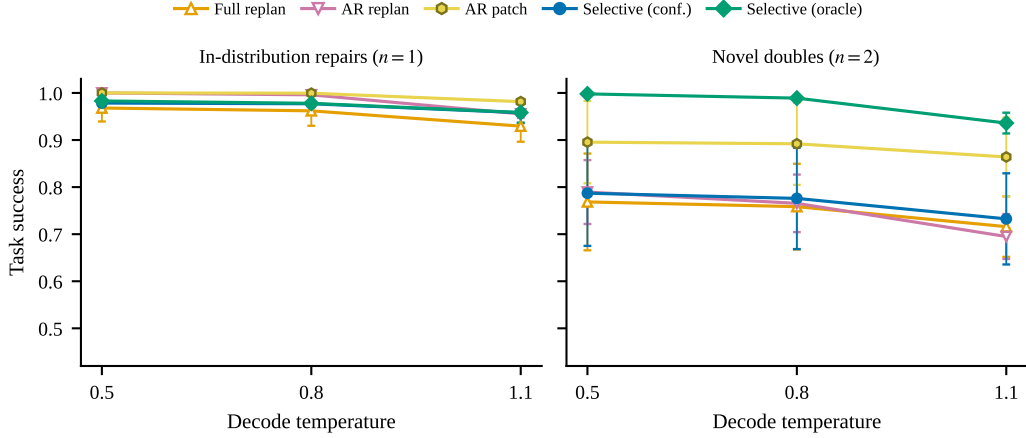


Figure 5: Decode-temperature robustness (5 seeds). Left: in-distribution repairs. Right: held-out doubles. Ordering is preserved and gaps widen with sampling noise.

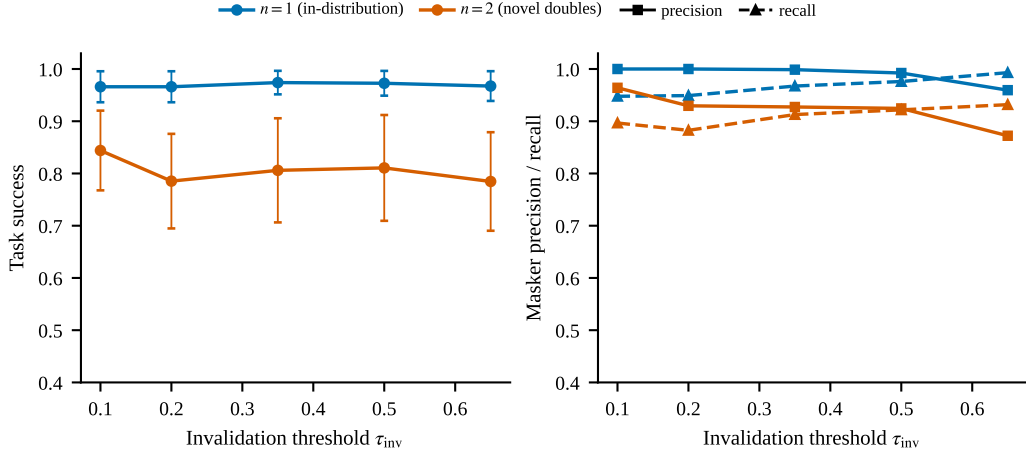


Figure 6: Masker operating curve (5 seeds). Left: task success vs. the invalidation threshold τ_{inv} at $n=1$ and on novel doubles. Right: masker precision (solid) and recall (dashed) against the oracle invalidation set. Success is flat across a $6.5\times$ threshold range at both n ; no operating point recovers random composition.

Derivation, and a correction. For the logits z_i at a masked position, the per-token gradient of the (naive) objective is $\partial \mathcal{L} / \partial z_i = (A_b / |S_b|) (p_i - e_{y_i})$, whose norm $\frac{|A_b|}{|S_b|} \|p_i - e_{y_i}\|_2 \leq \frac{|A_b|}{|S_b|} \sqrt{2} (1 - p_i(y_i))$ vanishes as $p_i(y_i) \rightarrow 1$ for either sign of A_b . An earlier draft attributed the collapse to a “saturation asymmetry” in which positive gradients vanish while negative ones stay large; at fixed confidence this is wrong, and we retract it. What is asymmetric is not the instantaneous gradient but the reachable effect (probabilities are bounded above, and a negative update redistributes the removed mass without specifying where it goes) — a candidate mechanism we do not claim to have isolated.

Controlled arms (seed 0, everything fixed except the advantage transform; Figure 8 left). $\text{lr } 10^{-4}$, $\beta=1$, ratio clipping on, 100 iterations: *filtered* $0.865 \rightarrow 0.979$; *anchor-only* ($A_b=0$, i.e. continued SFT) $0.865 \rightarrow 0.979$ — indistinguishable from the full recipe over this horizon, so **the gains are dominated by continued supervised learning**; *signed* $\rightarrow 0.969$ (the strong anchor prevents the v1 collapse); *random-sign* (± 1 , zero mean) $\rightarrow 0.938$, so gradient noise alone degrades little; *negative-only* clip($\hat{A}, -1.5, 0$) collapses to 0.083 by iteration 60 *despite* the full anchor and ratio clipping. The destructive component is the systematic negative direction, not variance and not

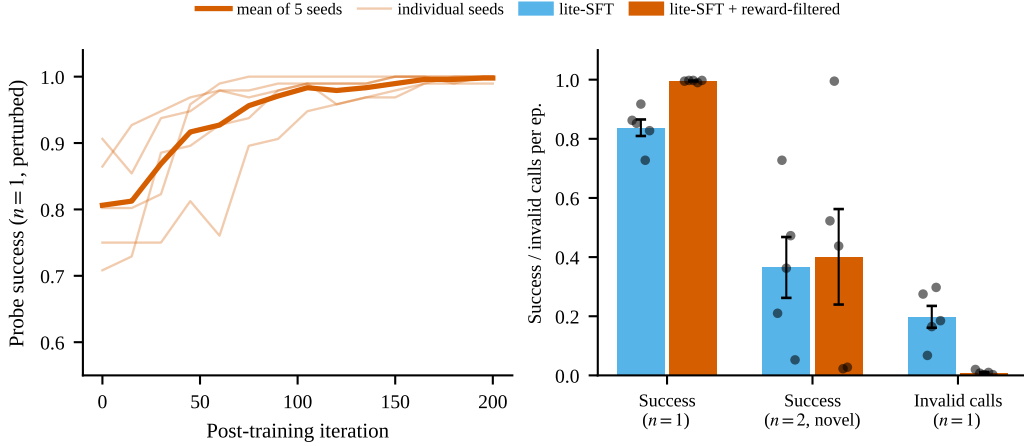


Figure 7: Reward-filtered imitation from a supervision-scarce SFT checkpoint (5 seeds). Left: probe success on perturbed episodes; all seeds reach 0.99–1.00. Right: lite-SFT vs. post-trained; in-distribution repair and invalid calls improve sharply, held-out doubles barely and with enormous seed variance. Dots are individual seeds. See Appendix C for the decomposition showing continued SFT explains most of the left panel.

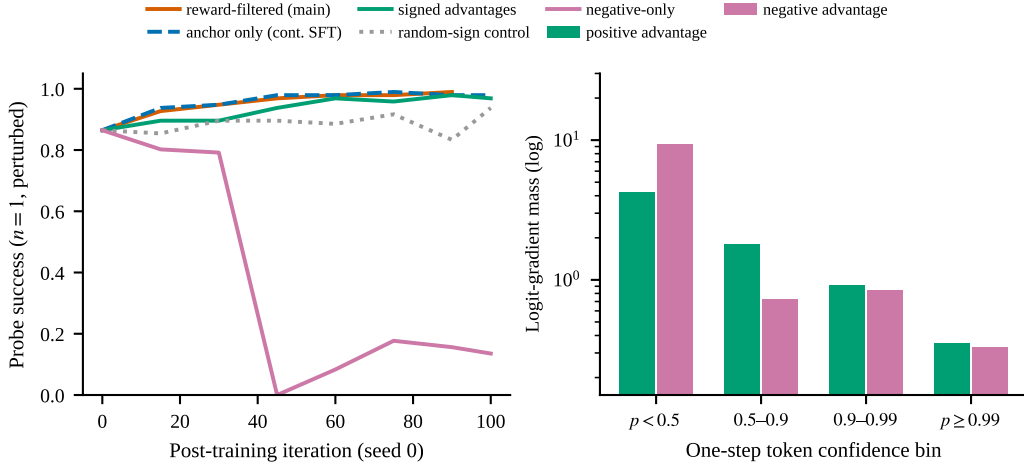


Figure 8: **Decomposing the post-training gains** (seed 0). Left: probe success under five advantage transforms with everything else fixed; anchor-only (continued SFT) matches the full recipe, and isolated negative advantages collapse the policy despite the anchor and ratio clipping. Right: exact per-token logit-gradient mass $|A_b| \|p - e_{y_i}\| / |S_b|$ by one-step confidence bin (log scale): mass at saturation is tiny and sign-symmetric; the negative skew is concentrated at $p < 0.5$.

clipping interactions.

Confidence and gradient-mass diagnostics (Figure 8 right). Over 480 training-distribution rollouts at the lite checkpoint, the one-step confidence of sampled tokens is nearly identical for successful and failed episodes (median 0.998 in both; 5.1% vs. 7.9% below 0.9), ruling out the hypothesis that failed rollouts simply carry lower-confidence, larger-gradient tokens. Binning the exact per-token gradient mass $|A_b| \|p_i - e_{y_i}\| / |S_b|$ by confidence: mass at $p \geq 0.99$ is tiny and sign-symmetric (0.35 vs. 0.33), as the derivation requires, while 75% of all mass sits at $p < 0.5$ with a $2.2\times$ negative skew (9.27 vs. 4.21). The collapse therefore proceeds through concentrated negative updates on the small uncertain-token minority, with interference across masking patterns and conditioning states as the plausible-but-unisolated pathway; the missing masking term of the exact policy gradient [Raajesh

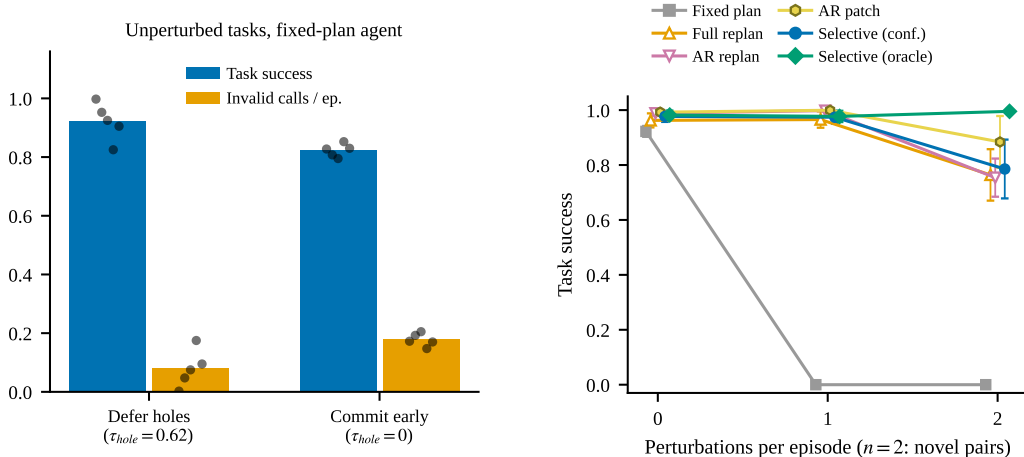


Figure 9: Left: typed-hole ablation on unperturbed episodes with the fixed-plan agent; deferring unresolvable tokens beats early commitment (5 seeds, dots per seed; one seed reverses). Right: success vs. number of perturbations for all agents (small x -offsets for visibility).

et al., 2026] is a further unmodeled source of misattributed credit.

Five seeds, and what the reward does buy. Under the final recipe, probe success rises from 0.71–0.91 to 0.99–1.00 on all five seeds (Figure 7); relative to anchor-only the reward term adds a small late-stage increment (to 1.000 by iteration 150 on seed 0) and drives invalid calls to 0.01. On novel doubles the post-trained policies score 0.401 ± 0.361 with bimodal per-seed outcomes (0.52/0.99/0.44/0.02/0.03) — on two seeds post-training *reduced* composition below the lite start while perfecting single-perturbation repair, consistent with filtering sharpening per-rule behaviors that then conflict under composition. A success-only behavior-cloning arm (RAFT proper) behaves like the filtered recipe, as expected given their similarity.

D The supervision-dose study

Design. The intervention adds k instances carrying *both* Family-A perturbations (suspended account + open invoice) with the dominance-correct expert repair (`escalate` at both reveals) to an otherwise unchanged training set; Family-C doubles remain fully held out, so the pure-composition test is preserved. Because inserting instances reshuffles the sampled data stream, dose arms are *not* “base + k examples”; the grid is therefore self-contained — all arms ($k \in \{0, 8, 64\}$) retrained through identical code at 1,500 steps, seeds 0–4 for $k \leq 8$ and 0–2 for $k=64$ — and all comparisons are within-grid. The grid uses full repair-snapshot coverage ($\text{cov} = 1.0$), for a reason worth reporting: an initial attempt at the base models’ 10% coverage produced one training draw whose repair conditionals for two perturbation kinds simply failed to form (single-perturbation success 0.15), a supervision-scarcity lottery that would have confounded the dose comparison. At $\text{cov} = 1.0$ the $k=0$ arm contains roughly ten times more single-perturbation repair supervision than the base models — a *stronger* control than the base configuration: if volume of primitive supervision could buy the priority rule, this arm would show it.

Primary endpoint. We probe the mechanism directly: mask the step-6 slot under the composed context and read $p(\text{escalate})$. Per seed: $k=0$: 0.42/0.02/0.13/0.47/1.00 (mean 0.408 ± 0.341 ; 1/5 above 0.9); $k=8$: 1.00/0.99/1.00/0.99/1.00 (0.997 ± 0.002 ; 5/5); $k=64$: ≥ 0.999 on all 3 seeds. Fisher exact on dominance-consistency (probe > 0.9), $k=8$ vs. $k=0$: one-sided $p = 0.024$. End-to-end Family-A doubles success: $0.589 \rightarrow 1.000 \rightarrow 1.000$; the $k=8$ arms are also clean on Family-C doubles, $n=1$, and $n=0$ (all ≥ 0.995 , five of five seeds). The same probe on the five *base* models gives 0.85/0.05/0.96/0.62/0.16, and its sign predicts each base seed’s Family-A outcome exactly.

Compute-only control. We warm-start the two dominance-failing base seeds (probe 0.05 and 0.16) from their final checkpoints and train for a second full budget (2,000 more steps, fresh optimizer and

schedule, base configuration otherwise). Both models remain healthy everywhere else (Family-C doubles, $n=1$, $n=0$ all 1.000) but Family-A doubles stay at 0.000 and the composed conditional *hardens* against the rule — $p(\text{escalate})$ falls to 0.000 and 0.001, with $p(\text{apply_credit}) \geq 0.998$. Doubling compute does not re-roll the coin; it entrenches the draw. This is the behavior one would expect if the composed context is simply attributed to whichever single-perturbation rule the model associated it with, and it is what makes the eight-demonstration result meaningful: the fix requires the rule to be *represented*, not merely more optimization against the same data.

Scale and data robustness. We repeat the grid at two further model sizes — $d=48$, 2 layers (46,144 parameters, $0.19\times$ the base width) and $d=144$, 3 layers (526,384 parameters, $2.1\times$) — and add a $4\times$ -data arm (16,000 instances at base width, dose 0), all under the grid configuration (Figure 10). The coin flip persists everywhere at dose 0: probe values 0.995/0.02/0.88 at 46k (1/3 dominance-consistent), 0.98/0.005/0.88 with $4\times$ data (1/3), and 0.27/0.18 at 526k (0/2; both draws lean toward the wrong rule), against 1/5 in the base-width grid — 3 of 13 dose-0 draws overall. Every healthy dose-8 draw is dominance-consistent: 3/3 at 46k, 5/5 at base width, 2/2 at 526k (probes ≥ 0.99 throughout), 10 of 10 pooled (exploratory one-sided Fisher across the heterogeneous configurations, $p = 2.5 \times 10^{-4}$). The threshold-crossing rule — end-to-end Family-A success iff the composed conditional clears τ_{inv} — holds on every healthy cell of the sweep. Three disclosures. (i) One 526k dose-8 draw suffered a mid-training loss spike (0.0075 $\rightarrow$ 0.16 at step 1,000, final 0.068) and is globally degraded ($n=1$ success 0.367); it is excluded from the tally, marked $\times$ in Figure 10, and was replaced by a fresh seed that converged normally. (ii) One 46k dose-8 draw exhibits the same Family-C conditional corruption as the base-grid dose-0 draws (Family-C success 0.000), and two 46k draws show mild $n=0$ dips (≈ 0.82): the earlier speculation that composed demonstrations regularize observation-conditioning does not survive the sweep — the corruption instability tracks capacity and supervision mix, not dose. (iii) All three $4\times$ -data draws are free of corruption pathologies while the composed conditional remains a lottery: data volume stabilizes the *supervised* conditionals and buys nothing for the *unsupervised* composed rule. The sweep spans 46k–526k parameters; it licenses a scale-trend claim within that range and nothing beyond it.

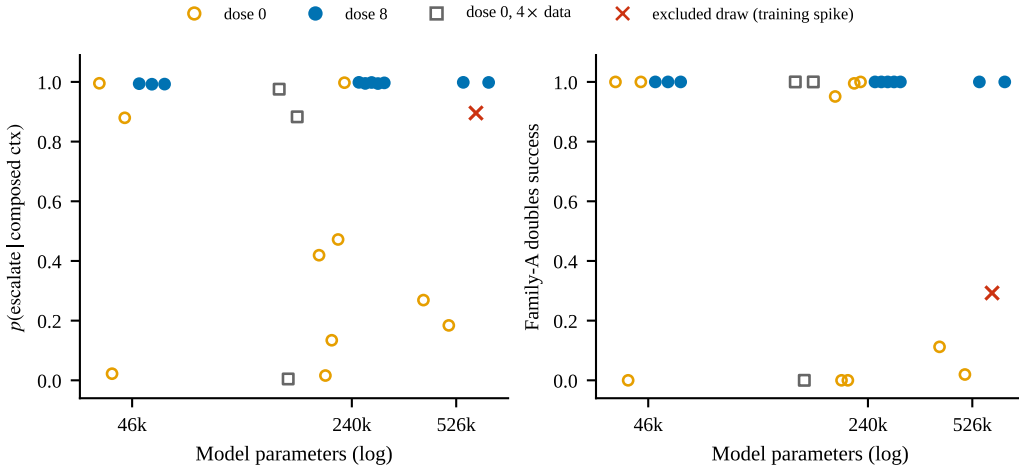


Figure 10: **Scale and data robustness of the dose effect.** Composed-context conditional (left) and Family-A doubles success (right) vs. model parameters (log), per seed. Open orange: dose 0; filled blue: dose 8; gray squares: dose 0 with $4\times$ training data (base width, x -dodged); red $\times$: the one excluded dose-8 draw with a documented training-loss spike. Dose-0 draws flip at every size and data volume; every healthy dose-8 draw clears the rule.

Side findings, reported as found. (1) Three of five $k=0$ draws corrupted an unrelated benign conditional: e.g. one seed rewrites `ship_label` $\rightarrow$ `swap_device` on *unperturbed* Family-C episodes (success 0.000), and this is not a threshold artifact — it persists at $\tau_{\text{inv}} = 0.15$, i.e. $p(\text{ship_label} |$

standard tier) itself collapsed, an over-repair instability of repair-heavy supervision mixes. (2) All five $k=8$ draws are free of these pathologies; the composed demonstrations may act as a regularizer on observation-conditioning itself, but with five draws we cannot exclude luck and flag that interpretation as speculative (the scale sweep below adds a low-capacity counterexample that further weakens it). (3) The grid differs from the base configuration (1,500 vs. 2,000 steps; cov 1.0 vs. 0.1), so dose-arm numbers are not directly comparable to Table 1; every claim above is within-grid.

E Reproduction

Training: 2,000 SFT steps, batch 64, AdamW with warmup-cosine (peak lr 3×10^{-3} , weight decay 10^{-4}); model $d=96$, 3 layers, 4 heads, ~ 250 k parameters. Decoding: temperature 0.7, 6 denoising rounds (4 on revision), $\tau_{\text{hole}}=0.62$, $\tau_{\text{inv}}=0.35$ (success and regeneration were flat across $\tau_{\text{inv}} \in \{0.25, 0.35, 0.5\}$ in a pilot; the full operating curve over $[0.10, 0.65]$ is Figure 6). Dose grid: identical training at 1,500 steps with cov = 1.0 and $k \in \{0, 8, 64\}$ forced A1+A2 instances; scale arms set model width/depth ($d \in \{48, 96, 144\}$) and dataset size (16,000 instances) via environment knobs. RL: 8 tasks $\times$ 6 rollouts per iteration, rollout temperature 1.0, advantages group-normalized then clipped to $[0, 1.5]$, ratio clip $\varepsilon=0.2$, replay anchor $\beta=1.0$, lr 10^{-4} , 200 iterations. Evaluation: 400 episodes per cell, 5 seeds ($n \leq 2$ conditions, temperature sweeps, hole ablation), 3 seeds for the RL condition. The full pipeline runs in ≈ 2.5 CPU-hours; all stages checkpoint and resume.